

University of Central Punjab
Faculty of Information Technology
LA Final Term Exam (2).

Course Instructors: Miss Fareeha Anjum, Dr. Maria Naseem, Miss Rohsha Tahir, Miss Maryam, Asad Zubair	Semester: Spring 2020
Course Code: CSSS2753	Marks: 20
Duration : 45 Minutes	Section:
Date : July 23 , 2020	Program: BSCS

Instructions:

1. Write your name, registration number, section and page numbering at each page of the answer sheet.
2. Make a single pdf file for uploading to hand in.
3. Before uploading the file, make sure to upload correct file, no excuse will be accepted if a wrong file is uploaded.
4. Attempt all questions.
5. All your calculations should be on your answer sheet.
6. You have first 5 minutes to read the paper carefully and then start solving the paper.

(5+4+1 marks)

Q1: (a) Determine whether the set $S = \{t^2 + 1, 3t^2 + 2t + 1, 6t^2 + 6t + 3\}$ span P^2 ?

(b) Determine whether the set S is linearly dependent or linearly independent?

(c) Does set S form basis for P^2 ?

(10 marks)

Q2: Find all positive Eigen values and corresponding Eigen vectors for

$$A = \begin{bmatrix} 0 & 10 & 0 \\ \frac{1}{10} & 0 & 0 \\ 0 & \frac{1}{5} & 0 \end{bmatrix}.$$